

Predicting Academic Success in Higher Education: A Machine Learning Approach

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Abstract

We address the prediction of academic success in higher education using a dataset of 4,424 students with demographic, socioeconomic, and semester-level performance information. Three tasks are developed: (1) **classification** of final academic status (Dropout/Enrolled/Graduate), where Gradient Boosting achieves $F1\text{-macro} = 0.716$ on 5-fold CV; (2) **regression** of the second-semester GPA (without data leakage), with $R^2 \approx 0.54$ and $RMSE \approx 0.95$; and (3) **K-Means clustering** ($k = 4$) identifying four student profiles with dropout rates between 9% and 80%. First-semester performance is the dominant predictor in both supervised tasks; socioeconomic variables such as payment status and debtor condition complement the risk profile.

Keywords: academic dropout prediction, gradient boosting, multi-class classification, GPA regression, K-Means clustering, early warning system, higher education, data leakage, class imbalance.

1 Introduction

Student dropout represents a critical challenge for higher education institutions worldwide, carrying both social and economic consequences. Early identification of at-risk students enables timely intervention and resource allocation. While prior work has addressed this problem using administrative data [1], comprehensive multi-task studies combining classification, regression, and unsupervised profiling remain underexplored.

We contribute a systematic analysis across three complementary machine learning tasks on a publicly available Portuguese university dataset. Key findings: (i) gradient boosting dominates classification; (ii) ensemble models capture strong nonlinearities between semesters; (iii) K-Means recovers interpretable risk-stratified profiles; and (iv) the Enrolled class is hardest to predict due to structural label ambiguity.

2 Dataset and Exploratory Analysis

The dataset contains **4,424 instances**, **36 variables**, and **no missing values**. Features cover demographics, socioeconomic context (scholarship, debtor status, tuition payment), prior qualifications, and enrollment/grade data for two semesters.

Class distribution. The target shows moderate imbalance: Graduate 49.9% (2,209), Dropout 32.1% (1,421), Enrolled 17.9% (794). This motivates F1-macro as the primary metric and `class_weight= 'balanced'` in classifiers.

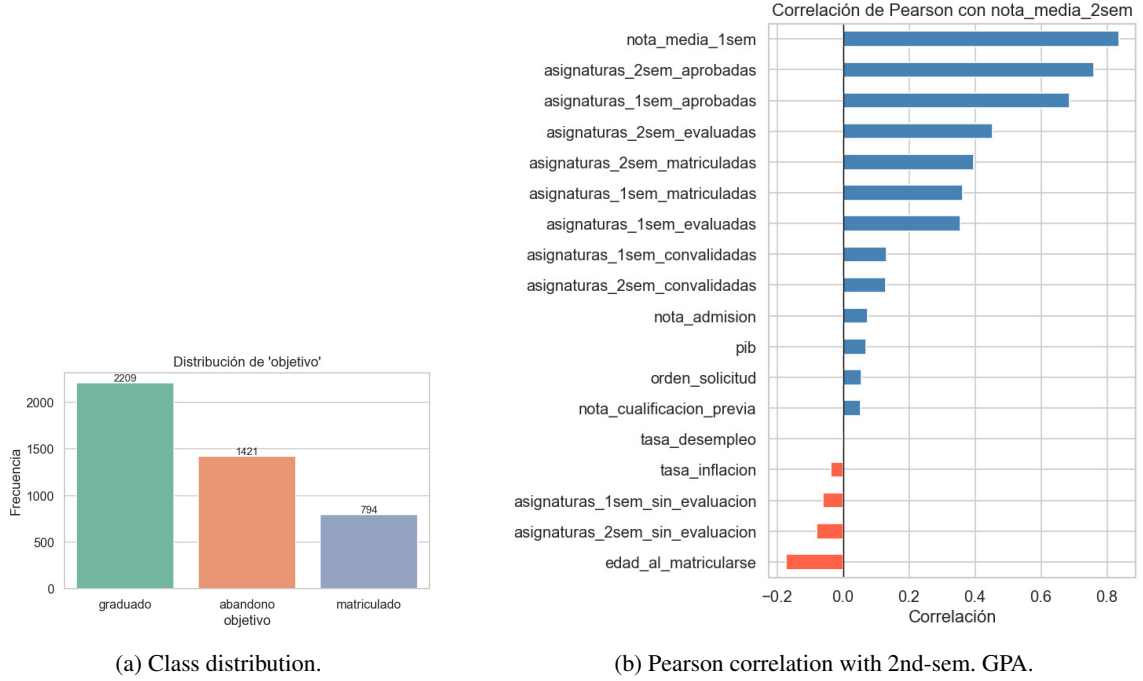


Figure 1: Class imbalance and feature correlations with the regression target. Variables from the 2nd semester with $r > 0.45$ are data leakage and excluded from regression.

Key findings. `nota_media_1sem` is the strongest predictor ($r = 0.82$). `matricula_al_dia` and `deudor` strongly discriminate dropout. Scholarship holders graduate more; evening students graduate less (consistent with a working-student profile).

3 Preprocessing and Methodology

Pipeline. All preprocessing is inside a `ColumnTransformer` to prevent leakage: `StandardScaler` for numeric features, `OneHotEncoder` for high-cardinality categoricals, 0/1 for binary variables. Stratified 80/20 split. `RANDOM_STATE = 42`.

Leakage in regression. The 2nd-semester variables contemporaneous with the target (`asignaturas_2sem_*`) are excluded. Including them would inflate R^2 spuriously ($r = 0.76$ between `asignaturas_2sem_aprobadas` and the target).

4 Task 1: Classification

Four classifiers evaluated via stratified 5-fold CV: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF, 300 trees), Gradient Boosting (GB, 200 iters, $lr = 0.05$).

Analysis. GB achieves the best F1-macro ($= 0.716$). LR is surprisingly competitive. RF scores lower F1-macro because unbalanced weights hurt the minority class. Applying `class_weight= 'balanced'` improves Enrolled recall from 37% to 56% at a moderate cost for Graduate.

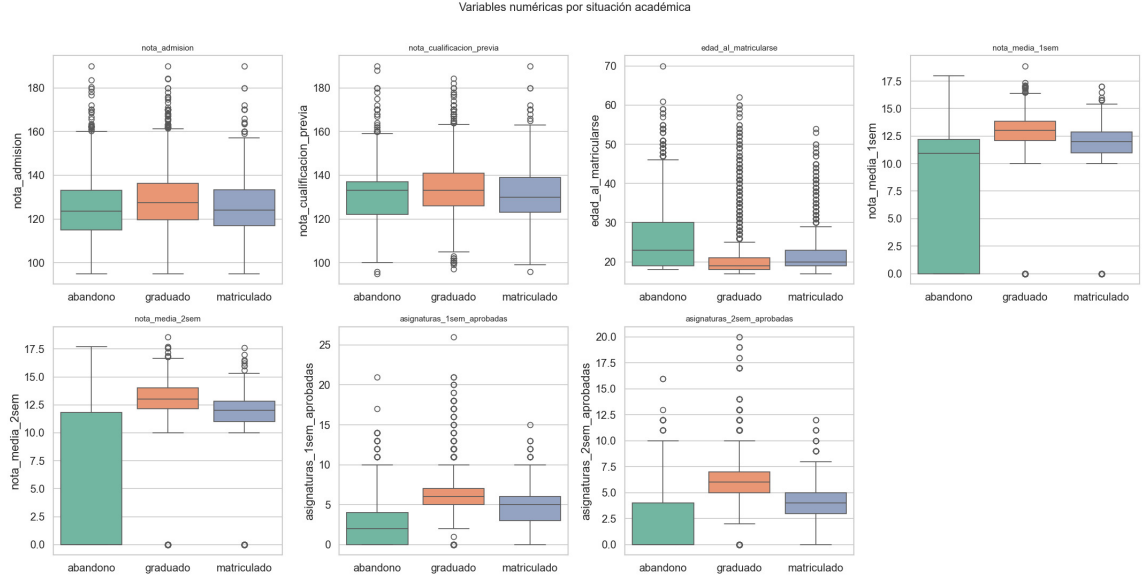


Figure 2: Key numeric variables by academic status. Graduates have higher 1st-semester grades and more approved courses. Dropout students are systematically older at enrollment.

Table 1: 5-fold CV classification results. **Bold**: best per column.

Model	Accuracy	F1-macro	F1-weighted
Logistic Regression	0.755	0.712	0.757
Decision Tree	0.703	0.668	0.718
Random Forest	0.779	0.690	0.756
Gradient Boosting	0.784	0.716	0.783

5 Task 2: Regression

We predict `nota_media_2sem` using only pre-2nd-semester information. Filtering to non-zero grades: mean = 12.73, std $\approx$ 1.4.

Table 2: 5-fold CV regression results.

Model	RMSE	MAE	R^2
Ridge	1.063	0.843	0.374
Lasso	1.050	0.831	0.390
Decision Tree	1.025	0.800	0.430
Random Forest	0.970	0.736	0.510
Gradient Boosting	0.946	0.719	0.537

Analysis. Ensembles outperform linear models ($\Delta R^2 \approx 0.16$), evidencing nonlinear inter-semester relationships. `nota_media_1sem` dominates importance (MDI $\approx$ 0.48). Mid-year predictions are feasible with enough lead time to activate support programs.

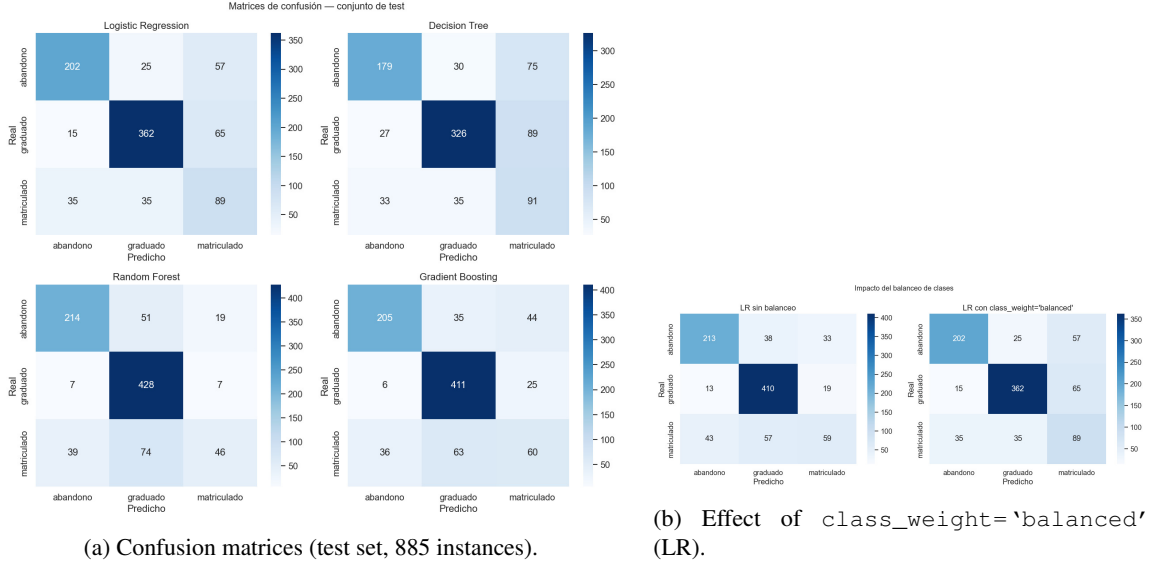


Figure 3: Test evaluation. Enrolled is the hardest class to predict across all models.

6 Task 3: Unsupervised Learning

Fifteen pre-outcome features were used: admission profile, socioeconomic status, 1st-semester performance, and macroeconomic context.

Dimensionality and k selection. PCA needs 8 components for 80% variance ($PC1=17\%$, $PC2=12.8\%$). Silhouette peaks at $k=4$ (score ≈ 0.223), with a sharp drop for $k \geq 5$.

Table 3: Characterization of the 4 clusters. Values in standard deviations.

Cluster	Profile	Main characteristics
0	Uncertain transition	Age +1.66 std, average grades. 43% dropout, 43% graduate.
1	Moderate performance	Admission +0.85 std, good 1st sem. 54% graduate, 21% dropout.
2	High performance	High admission & prior qual., no debt (+1.72 std). 80% graduate.
3	High risk	Admission -1.61 std, 1st-sem -1.67 std, high debt +1.72 std. 80% dropout.

7 Discussion

Convergent risk factors. All three analyses agree: dropout is associated with low initial academic performance, precarious economic situation, and older enrollment age. Cluster 3 concentrates all these factors simultaneously (80% dropout rate), making it the primary target for institutional intervention.

Actionable interventions. Two categories of intervention emerge. First, *economic support*: targeting debtors and students with overdue tuition for financial aid before academic damage accumulates. Second, *academic support*: intensive tutoring and mentoring in the first semester, directly addressing the dominant predictor. Both levers are identifiable from information available at mid-first-year.

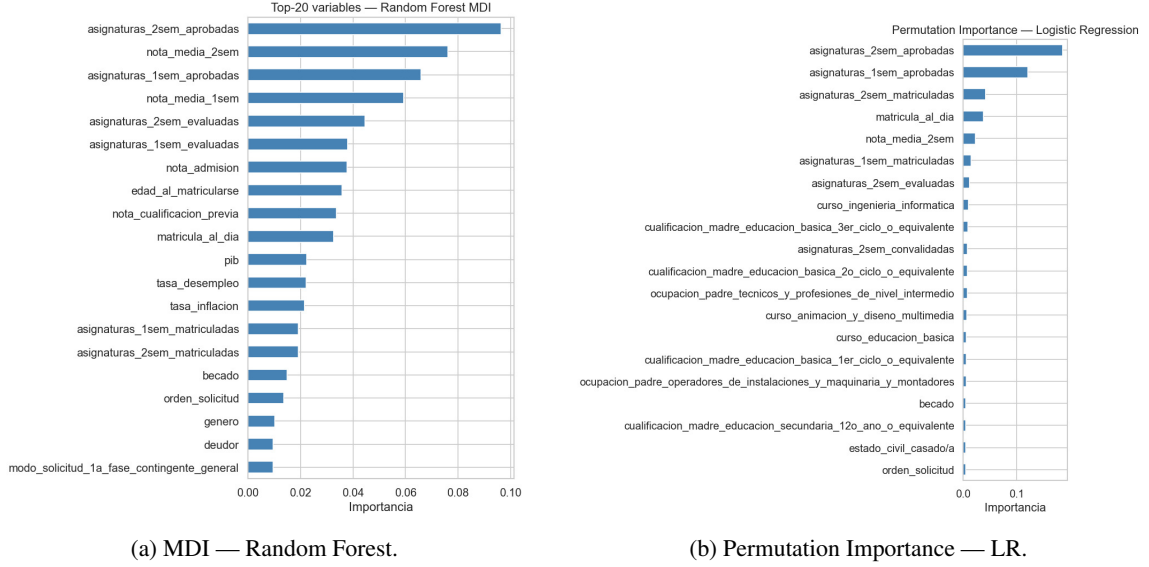


Figure 4: Feature importance in classification. Both metrics highlight semester performance, admission grades, and payment status. 2nd-semester features dominate MDI but should be excluded in a real early-warning deployment.

Early warning potential. Both the classifier ($F1\text{-macro} = 0.716$, accuracy $\approx 78\%$) and the regressor ($R^2 = 0.54$, $RMSE = 0.95$) are operational from the middle of the first academic year, making real-time risk flagging feasible without requiring end-of-year outcomes. Cluster 3 additionally provides an unsupervised, label-free mechanism for the same purpose.

Limitations. (i) Correlation $\neq$ causation: low grades may reflect unmeasured external factors not captured in the dataset. (ii) Single Portuguese institution: generalization to other educational contexts and countries requires external validation. (iii) Structural label ambiguity: the Enrolled class conflates future graduates and future dropouts, creating irreducible prediction difficulty (recall $\approx 38\%$ across all models).

8 Conclusion

We presented a comprehensive multi-task machine learning study on academic success prediction in higher education, covering classification, regression, and unsupervised clustering on a real-world dataset of 4,424 students.

Gradient Boosting consistently outperforms competing models in both supervised tasks, achieving $F1\text{-macro} = 0.716$ for three-class classification and $R^2 = 0.54$ for second-semester GPA regression without data leakage. The first-semester GPA ($r = 0.82$) is the single dominant predictor across all analyses, with socioeconomic indicators—debtor status and tuition payment—providing complementary and actionable risk signals.

K-Means clustering ($k = 4$) identifies a high-risk profile (Cluster 3, 80% dropout) characterized by low admission grades, poor first-semester performance, and high debt load. These characteristics are directly addressable through institutional programs. Both unsupervised and supervised models are deployable at mid-first-year, making an early-warning system operationally feasible with meaningful predictive power well before final outcomes are known.

Future work should explore: (i) transfer learning across institutions to improve generalizability; (ii) temporal modeling of semester-by-semester progression; and (iii) fairness audits to ensure interventions do not inadvertently discriminate based on demographic attributes.

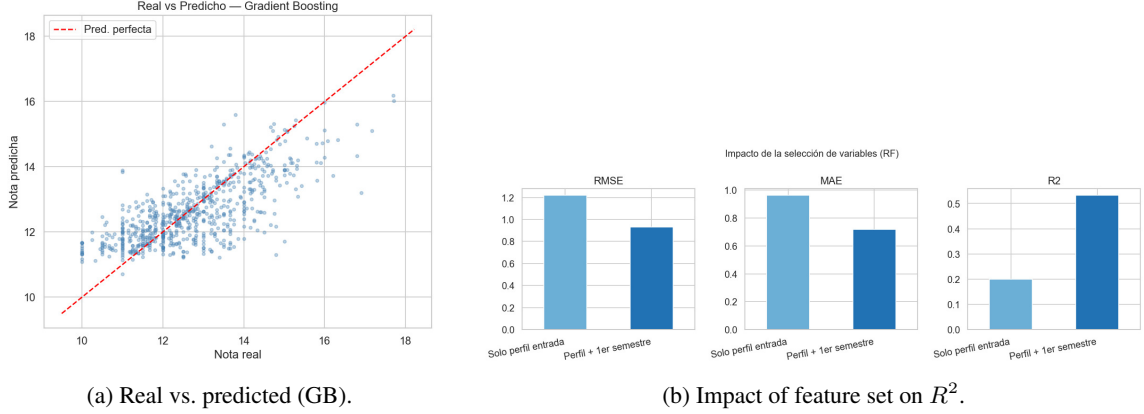


Figure 5: Regression diagnostics. No systematic bias in residuals. Adding 1st-semester information improves R^2 by 169% ($0.20 \rightarrow 0.54$).

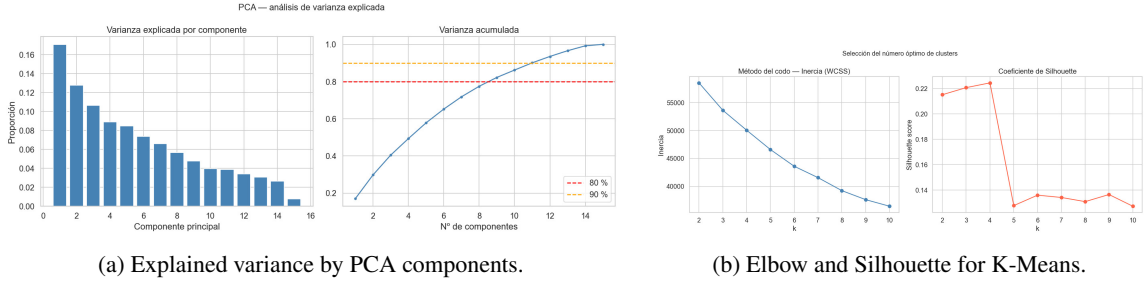


Figure 6: PCA requires 8 components for 80% variance. Silhouette confirms $k = 4$.

References

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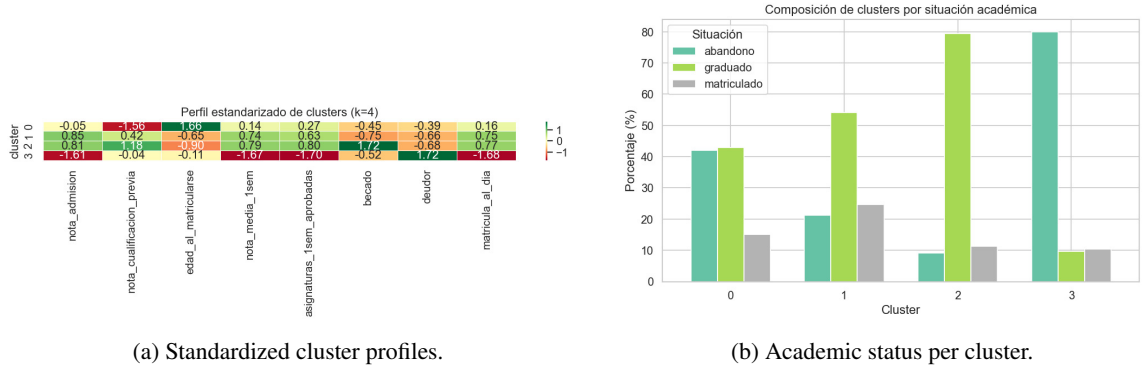


Figure 7: Cluster 3 (high risk): 80% dropout. Cluster 2 (high performance): 80% graduate.

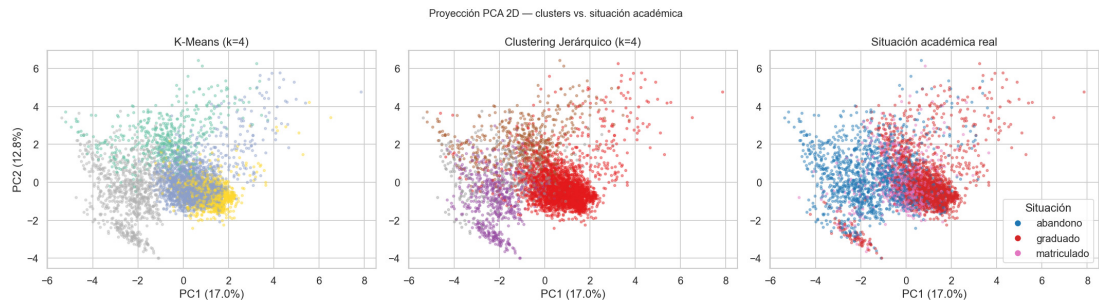


Figure 8: 2D PCA projection: K-Means (left), hierarchical clustering (centre), real academic status (right). Both methods produce similar partitions, confirming cluster robustness.